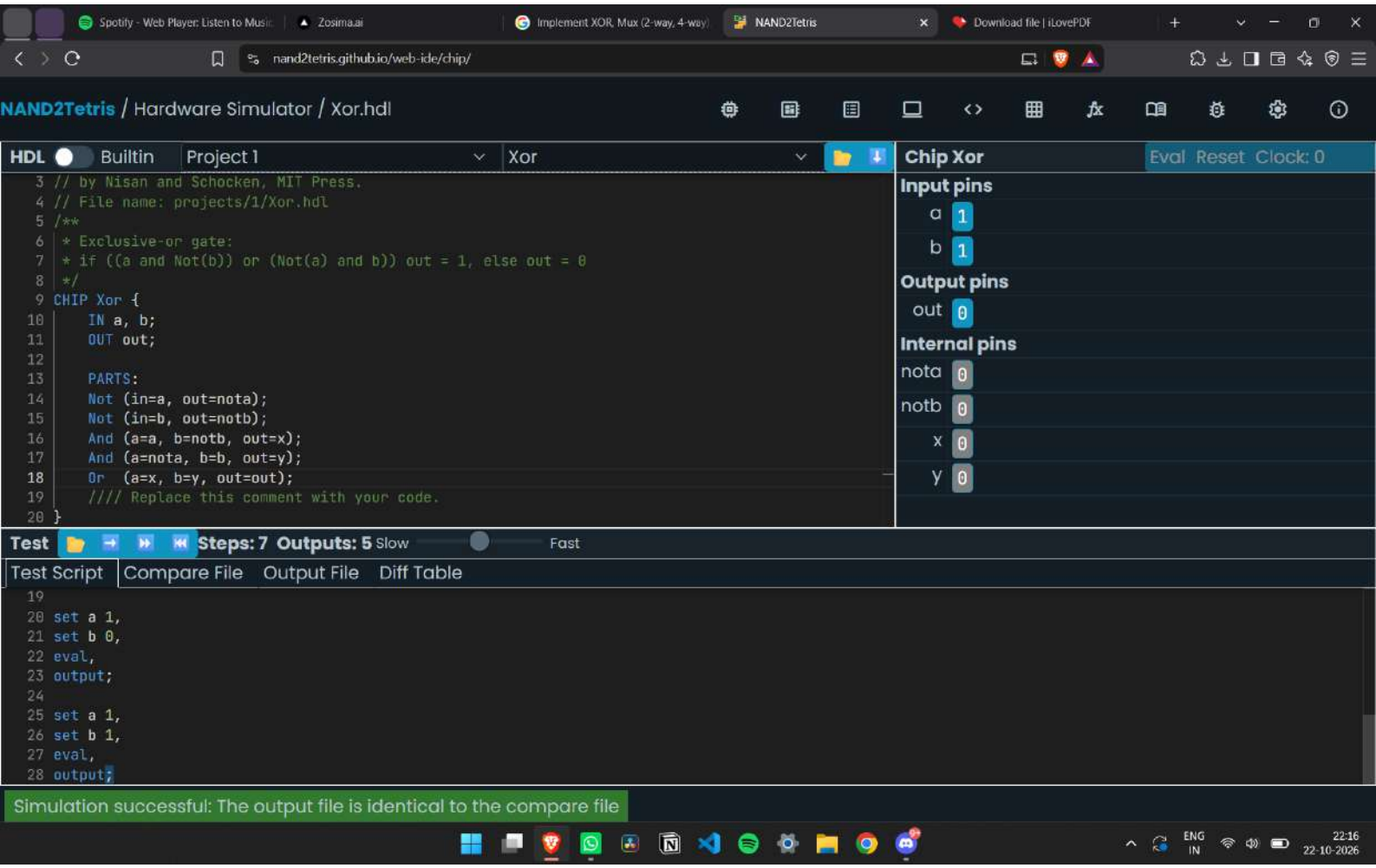
Spotify - Web Player: Listen to Music
Zosima.ai
Implement XOR, Mux (2-way, 4-way)
NAND2Tetris
Download file | iLovePDF
nand2tetris.github.io/web-ide/chip/
NAND2Tetris / Hardware Simulator / Xor.hdl
HDL
Builtin
Project 1
Xor
// by Nisan and Schocken, MIT Press.
// File name: projects/1/Xor.hdl
/**
* Exclusive-or gate:
* if ((a and Not(b)) or (Not(a) and b)) out = 1, else out = 0
*/
CHIP Xor {
IN a, b;
OUT out;
PARTS:
Not (in=a, out=nota);
Not (in=b, out=notb);
And (a=a, b=notb, out=x);
And (a=nota, b=b, out=y);
Or (a=x, b=y, out=out);
//// Replace this comment with your code.
}
Chip Xor
Eval Reset Clock: 0
Input pins
a 1
b 1
Output pins
out 0
Internal pins
nota 0
notb 0
x 0
y 0
Test
Steps: 7 Outputs: 5 Slow
Fast
Test Script
Compare File
Output File
Diff Table
set a 1,
set b 0,
eval,
output;
set a 1,
set b 1,
eval,
output;
Simulation successful: The output file is identical to the compare file
ENG
IN
22:16
22-10-2026